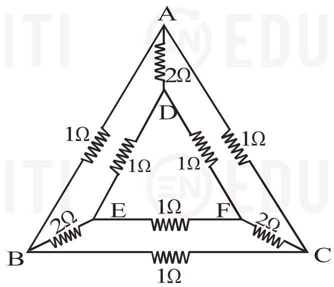
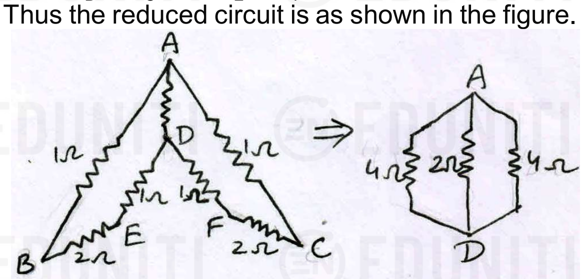
1. A network of nine conductors connects six points A, B, C, D, E and F as shown in figure. The figure denotes resistances in ohms. Find the equivalents resistance between A and D.



1. From the symmetry of the circuit about AD we can see that $V_B = V_C$ and $V_E = V_F$

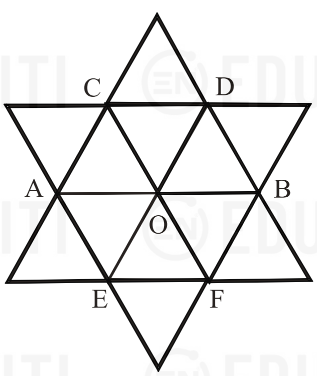
Thus the reduced circuit is as shown in the figure.



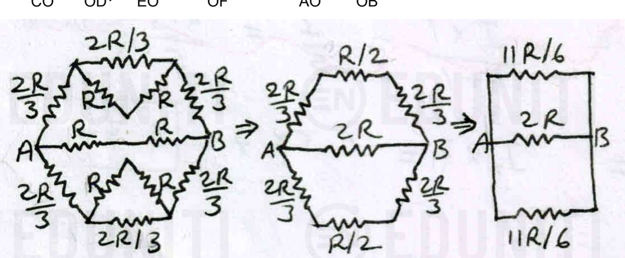
The equivalent resistance across AD is given by

$$\frac{1}{R_{AD}} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 \qquad \therefore \boxed{R_{AD} = 1\Omega}$$

2. Find the equivalent resistance of the circuit between points A and B shown in figure. The resistance of each branch is R



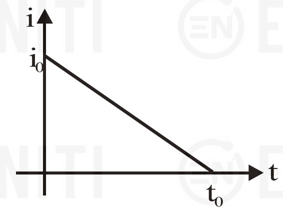
2. Due to symmetry, we observe that $I_{CO} = I_{OD}$, $I_{EO} = E_{OF}$ and $I_{AO} = I_{OB}$.



We can therefore detatch the branches at centre O and redraw the circuit as shown in figure. The equivalent resistance across AB is given by

$$\frac{1}{R_{AB}} = \frac{6}{11R} + \frac{1}{2R} + \frac{6}{11R} = \frac{35}{22R} \qquad \therefore \boxed{R_{AB} = \frac{22}{35}R}$$

3. Relation between current in conductor and time is shown in figure.



(a) Find the total charge flow through the conductor

(b) Write the expression of current in terms of time

(c) If the resistance of conductor is R, then the total heat dissipated across resistance R is

3.(a) The total charge flow q is equal to the area under the given $i-t$ graph.

$$\therefore q = \frac{1}{2} \times i_0 \times t_0 = \boxed{\frac{i_0 t_0}{2}}$$

(b) This graph is of form, y = *mx* + *c*

where, $m = -i_0/t_0$ and $c = i_0$

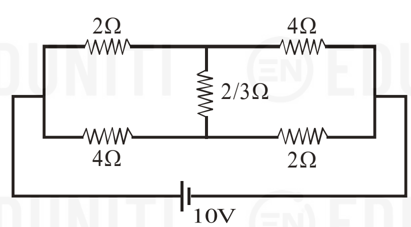
Therefore, the expression of current in terms of time is

$$i = \frac{-i_0}{t_0}t + i_0 \qquad \therefore \boxed{i = i_0\left(1 - \frac{t}{t_0}\right)}$$

(c) $$H = \int i^2 R dt = i_0^2 R \int_0^{t_0} \left(1 - \frac{t}{t_0}\right)^2 dt = i_0^2 R \int_0^{t_0} \left(1 - \frac{2t}{t_0} + \frac{t^2}{t_0^2}\right) dt$$

$$= \boxed{\frac{i_0^2 R t_0}{3}}$$

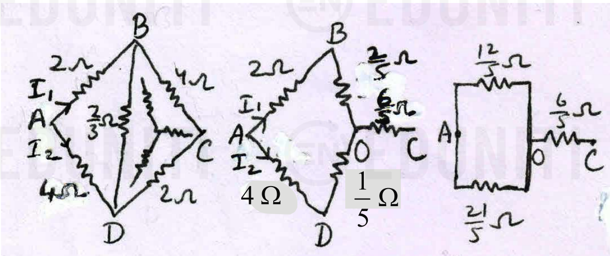
4. Find the current through $(2/3)\Omega$ resistor in the figure shown.



4. Converting the delta *BCD* into star, we get the resistor network as shown in the fig.

The resistances of *BD*, *BC* and *CD* are respectively $(2/3)\Omega$, 4Ω and 2Ω. Their sum is $\frac{2}{3} + 4 + 2 = \frac{20}{3}\Omega$.

So, resistances BO, OD and OC are respectively



$$\frac{(2/3)\times 4}{20/3}=\frac{2}{5}\Omega,\quad \frac{(2/3)\times 2}{20/3}=\frac{1}{5}\Omega \text{ and } \frac{2\times 4}{20/3}=\frac{6}{5}\Omega$$

Now, $R_{AO}=\dfrac{12/5\times 21/5}{12/5+21/5}=\dfrac{84}{55}\Omega$ and $R_{OC}=\dfrac{6}{5}\Omega$

$$\frac{V_{AO}}{V_{OC}}=\frac{R_{AO}}{R_{OC}}=\frac{84/55}{6/5}=\frac{14}{11}$$

$$\Rightarrow V_{AO}=\frac{14}{14+11}\times 10=\frac{28}{5}V$$

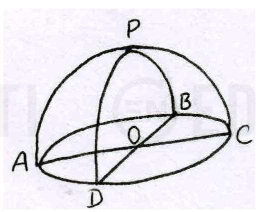
$$\Rightarrow I_1=\frac{V_{AO}}{12/5}=\frac{28/5}{12/5}=\frac{7}{3}A \text{ and}$$

$$I_2=\frac{V_{AO}}{21/5}=\frac{28/5}{21/5}=\frac{4}{3}A$$

Therefore current through $(2/3)\Omega$ resistor is

$$I_1-I_2=7/3-4/3=\boxed{1A}$$

5. A hemisphere network of radius 'a' is made by using a conducting wire of resistance per unit length r. Find the equivalent resistance across OP.

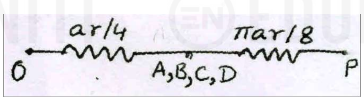


5. Here, $R_{OA}=R_{OB}=R_{OC}=R_{OD}=ar$

and $R_{PA}=R_{PB}=R_{PC}=R_{PD}=\dfrac{\pi}{2}ar$

From symmetry, $V_A=V_B=V_C=V_D$

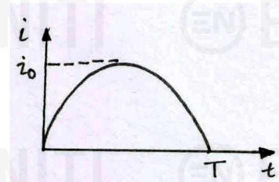
So, the circuit can be drawn as shown in the figure.



$$R_{OP}=\frac{ar}{4}+\frac{\pi ar}{8}=\boxed{\frac{ar}{8}(2+\pi)}$$

6. A total charge Q flows across a resistor R during a time interval T in such a way that the current v/s time graph for 0 to T is like the loop of a sine curve in the range 0 to π. What will be the total heat generated in the resistor.

6. Since the variation of i vs t for t = 0 to T is a sine curve in the range 0 to π, we have



$i=i_0\sin\left(\dfrac{\pi t}{T}\right)$, where i_0 is peak current.

$$\Rightarrow Q=\int_0^T idt=i_0\int_0^T\sin\left(\frac{\pi t}{T}\right)dt=\frac{i_0T}{\pi}\cos\left(\frac{\pi t}{T}\right)\Bigg|_T^0=\frac{2i_0T}{\pi}$$

$$\Rightarrow i_0=\frac{\pi Q}{2T}$$

Heat generated in resistor R during time t = 0 to T will be

$$H=\int_0^T i^2Rdt=i_0^2R\int_0^T\sin^2\left(\frac{\pi t}{T}\right)dt=\frac{i_0^2R}{2}\int_0^T\left(1-\cos\frac{2\pi t}{T}\right)dt$$

$$=\frac{i_0^2R}{2}\left[t-\frac{\sin(2\pi t/T)}{2\pi/T}\right]_0^T=\frac{i_0^2RT}{2}$$

$$\therefore H=\left(\frac{\pi Q}{2T}\right)^2\times\frac{RT}{2}=\boxed{\frac{\pi^2Q^2R}{8T}}$$

7. A rod of length L and cross–section area A lies along the x–axis between x = 0 and x = L. The material obeys ohm's law and its resistivity varies along the rod according to $\rho(x)=\rho_0e^{-x/L}$. The potential is V_0 at x = 0 and is zero at x = L.

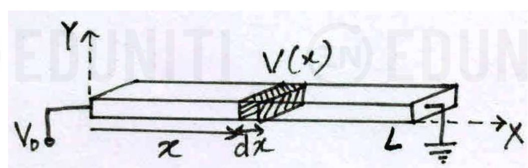
(a) Find the total resistance of the rod and the current in the rod.

(b) Find the electric potential in the rod as a function of x.

7.(a) Small resistance of an element of thickness dx is,

$$dR=\frac{\rho dx}{A}$$

The resistance of the rod from 0 to x is



$$R(x)=\int_0^x\frac{\rho_0}{A}e^{-\frac{x}{L}}dx=\frac{\rho_0L}{A}\left[e^{-x/L}\right]_x^0=\frac{\rho_0L}{A}(1-e^{-x/L})$$

The total resistance of rod is

$$R=R(L)=\frac{\rho_0L}{A}(1-e^{-L/L})=\boxed{\frac{\rho_0L}{A}\left(1-\frac{1}{e}\right)}$$

The current through rod is $I=\dfrac{V_0}{R}=\boxed{\dfrac{V_0Ae}{\rho_0L(e-1)}}$

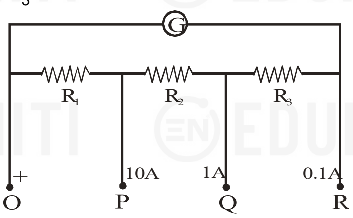
(b) If V(x) is the potential at x, then

$$V_0-V(x)=IR(x)=\frac{V_0Ae}{\rho_0L(e-1)}\times\frac{\rho_0L}{A}(1-e^{-x/L})$$

$$=\frac{V_0(e-e^{1-x/L})}{e-1}$$

$\therefore\ V(x) = V_0 - \dfrac{V_0(e - e^{1-x/L})}{e-1} = \boxed{\dfrac{V_0(e^{1-x/L}-1)}{e-1}}$

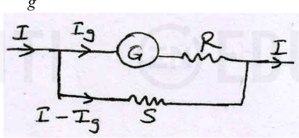
8. The resistance of the galvanometer G in the circuit is $25\,\Omega$. The meter deflects full scale for a current of 10 mA. The meter behaves as an ammeter of three different ranges. The range is 0–10A, if the terminals O and P are taken, the range is 0–1 A between O and Q and the range is 0–0.1 A between O and R. Calculate the resistances R_1, R_2 and R_3.



8. Here, $G = 25\Omega$ and $I_g = 0.01A$

In the figure shown, $(I - I_g)S = I_g(G+R)$

$\Rightarrow \dfrac{G+R}{S} = \dfrac{I - I_g}{I_g}$



When O and P are considered, $R = R_2 + R_3$ and $S = R_1$.

So, $\dfrac{25 + R_2 + R_3}{R_1} = \dfrac{10 - 0.01}{0.01} = 999$

$\Rightarrow 999R_1 - R_2 - R_3 = 999$ .....(1)

For the other two cases,

$\dfrac{25 + R_3}{R_1 + R_2} = \dfrac{1 - 0.01}{0.01} = 99$

and $\dfrac{25}{R_1 + R_2 + R_3} = \dfrac{0.1 - 0.01}{0.01} = 9$

$\Rightarrow 99R_1 + 99R_2 - R_3 = 25$ ....(2)

and $9R_1 + 9R_2 + 9R_3 = 25$ .....(3)

From (1) and (2), $900R_1 - 100R_2 = 0$ .....(4)

From (2) and (3), $900R_1 + 900R_2 = 250$ .....(5)

From (4) and (5), $\boxed{R_1 = \dfrac{1}{36}\Omega}$ and $\boxed{R_2 = \dfrac{1}{4}\Omega}$

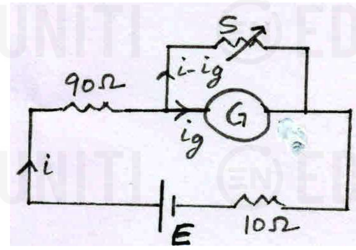
On putting these in (3), we get $\boxed{R_3 = 2.5\Omega}$

9. A galvanometer having 50 divisions and full scale deflection current 300 mA is provided with a variable shunt. It is used to measure the current as a ammeter when connected in series with a resistance of 90 Ω and a battery of internal resistance $10\,\Omega$. It is observed that when the shunt resistances are $10\,\Omega$ and $50\,\Omega$ respectively, the deflections are respectively 9 and 30 divisions. What is the resistance of the galvanometer and the emf of the cell?

9. The galvanometer has 50 divisions and full scale deflection current is 0.3A. So, current for a deflection of 1 unit is $\mu = 0.3/50 = 0.006\,A/div.$

Given, for $S = 10\Omega$, $i_g = 9\mu$ and for $S = 50\Omega$, $i_g = 30\mu$.

The equivalent resistance of the circuit is



$R_{eq} = \dfrac{GS}{G+S} + 90 + 10 = \dfrac{GS + 100(G+S)}{G+S}$

$\Rightarrow i = \dfrac{E}{R_{eq}}$ and

$i_g = \dfrac{S}{G+S}i = \dfrac{S}{G+S} \times \dfrac{E(G+S)}{GS + 100(G+S)} = \dfrac{ES}{GS + 100(G+S)}$

For, $S = 10\Omega$, $i_g = 9\mu = 9 \times 0.006A$

$\Rightarrow 9 \times 0.006 = \dfrac{10E}{10G + 100(G+10)} = \dfrac{E}{11G + 100}$ .....(1)

For, $S = 50\Omega$, $i_g = 30\mu$

$\Rightarrow 30 \times 0.006 = \dfrac{50E}{50G + 100(G+50)} = \dfrac{E}{3G+100}$ .....(2)

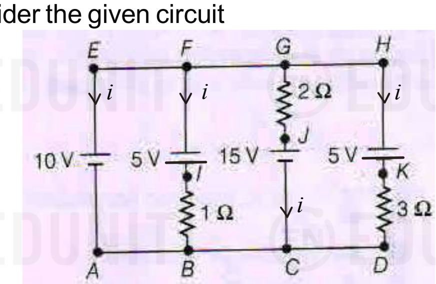
From (1) and (2), we get $\dfrac{9}{30} = \dfrac{3G+100}{11G+100}$

$\therefore \boxed{G = \dfrac{700}{3}\Omega}$

On putting this in (2), we get $\boxed{E = 144V}$

Passage (Q 10 to 12)

Consider the given circuit



It is known that potential at point A is zero. Now, answer the following questions.

10. In the given circuit,
(a) one battery acts like source
(b) two batteries acts like source
(c) three batteries acts like source
(d) all batteries act like source

11. Which resistor consumes maximum power?
(a) $1\,\Omega$ resistor (b) $2\,\Omega$ resistor
(c) $3\,\Omega$ resistor (d) none of the above

12. Which battery consumes or gives maximum power?
(a) 10 V battery
(b) 15 V battery
(c) 5 V battery between points F and B
(d) 5 V battery between points H and D

Passage 10 to 12 (Solution)

10.(d) Here, $V_A = V_B = V_C = V_D = 0$

$\Rightarrow V_E = V_F = V_G = V_H = V_A + 10 = 10V$

$V_I = V_F + 5 = 15V,\ V_J = V_C + 15 = 15V,\ V_K = V_H + 5 = 15V$

$i_2 = (V_I + V_B)/1 = (15-0)/1 = 15A,$

$i_3 = (V_G - V_J)/2 = (10-15)/2 = -2.5A,$

$i_4 = (V_K - V_D)/3 = (15-0)/3 = 5A$

Applying kirchhoff's junction rule, $i_1 + i_2 + i_3 + i_4 = 0$

$\Rightarrow\ i_1 = -(i_2 + i_3 + i_4) = -(15 - 2.5 + 5) = -17.5A$

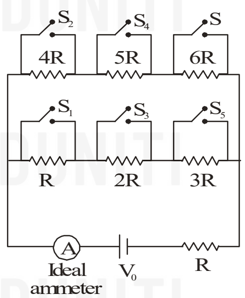
As $i_1 < 0$, 10V battery acts as source.
As $i_2 > 0$ and $i_4 > 0$, both 5V batteries act as source.
As $i_2 < 0$, 15V battery acts as source.
All batteries act as source.

11(a) Power consumed, $P = i^2 R$.
On solving for each resistor, we find that power consumed is maximum by 1Ω resistor.

12.(a) Power delivered by battery, $P' = EI$.
On solving for all batteries we find that power delivered is maximum by 10V battery.

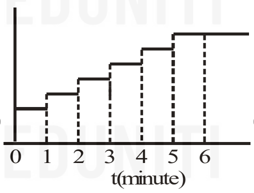
Passage 13 to 15

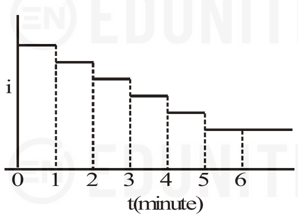
For the circuit shown, answer the following questions

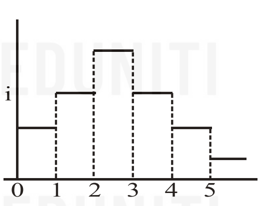


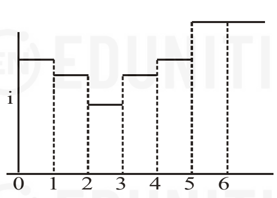
13. In which of the following case, the current shown by ammeter is maximum
(a) S_1, S_2, S_3 closed
(b) S_2, S_4, S_5 closed
(c) S_1, S_3, S_5 closed
(d) S_2, S_3, S_4 closed

14. If switches S_1, S_2 and so on upto S_6 are closed at regular intervals of 1minute starting from t = 0, the graph of current versus time is best represented as

(a)


(b)


(c)


(d)


15. Ratio of power developed by battery when all switches are closed to that when all switches are open

(a) $\frac{37}{7}$ (b) $\frac{7}{37}$
(c) $\left(\frac{37}{7}\right)^2$ (d) $\left(\frac{7}{37}\right)^2$

Passage Solution (13 to 15)

13.(c) When S_1, S_3 and S_5 are closed, equivalent resistance of the circuit is minimum and i.e., *R*. So, the current shown by ammeter is maximum.

14.(a) When a switch is closed, the resistance across it is short-circuited. So, with every switch closed, equivalent resistance of the circuit decreases. Thus, current increases at every step.

15.(a) When all switch are closed, the equivalent resistance,

$R_1 = R$

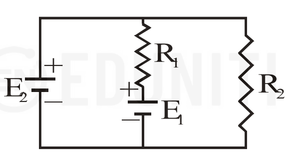
When all switches are open, the equivalent resistance,

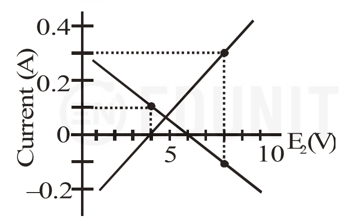
$$R_2 = \frac{6R \times 15R}{6R + 15R} + R = \frac{30R}{7} + R = \frac{37R}{7}$$

As $P = \frac{V_0^2}{R}$, we have $\frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{37R/7}{R} = \frac{37}{7}$

Passage 16 to 18

In the circuit shown, both batteries are ideal. Emf E_1 of battery 1 has a fixed value, but emf E_2 of battery 2 can be varied between 1 V and 10V. The graph gives the currents through the two batteries as a function of E_2, but are not marked as to which plot corresponds to which battery. But for both plots, current is assumed to be negative when the battery is being charged.





16. The value of emf E_1 is
(a) 8V (b) 6V
(c) 4V (d) 2V

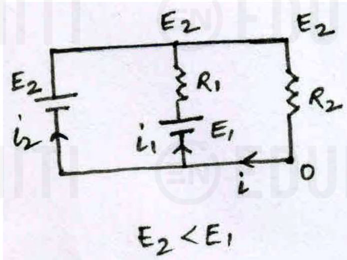
17. The resistance R_1 has value
(a) $10\ \Omega$ (b) $20\ \Omega$
(c) $30\ \Omega$ (d) $40\ \Omega$

18. The resistance R_2 is equal to
(a) $10\ \Omega$ (b) $20\ \Omega$
(c) $30\ \Omega$ (d) $40\ \Omega$

Passage 16 to 18 Solution

16.(b) For $E_2 < E_1$, i_1 will be positive. Thus graph with negative slope is of i_1.

$$i_1 = \frac{E_1 - E_2}{R_1} \quad .....(1)$$



From graph, $i_1 = 0$ when $E_2 = 6V$ $\therefore E_1 = 6V$.

17.(b)From graph, i_1= 0.1 A when E_2 = 4V
On putting these values in eqn (1), we get

$0.1 = \frac{6-4}{R_1}$ $\therefore R_1 = 20\Omega$

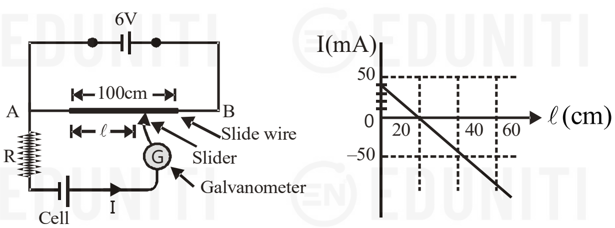
18.(d) From figure, when $i_2 = 0$, $i_1 = i$ and $i_1 = \frac{E_1}{R_1 + R_2}$

From graph, when $i_2 = 0$, $i_1 = 0.1$A

$\Rightarrow 0.1 = \frac{6}{20 + R_2}$ $\therefore R_2 = 40\Omega.$

Passage 19 to 22

In the circuit shown, the internal resistance of the cell is negligible. The distance of the jockey (slider) from left hand end of the wire is l. The adjoining graph shows the variation of current I (marked in figure) with length l of the slide wire.



19. For balance condition of the instrument, the value of l is equal to
(a) 40 cm (b) 20 cm
(c) 100 cm (d) None of these

20. The emf of cell is
(a) 0.98 V (b) 1.20 V
(c) 1.86 V (d) 3 V

21. Value of the resistance R is
(a) 30 Ω (b) 40 Ω
(c) 38 Ω (d) 45 Ω

22. The magnitude of current I when l = 100 cm is
(a) 50 mA (b) 120 mA
(c) 160 mA (d) 300 mA

Passage solution (19 to 22)

19.(b) From graph, I = 0 at l = 20 cm
So, for balanced condition, l = 20 cm.

20.(b) Let ε be emf of cell. In balanced condition, $\frac{\varepsilon}{20} = \frac{6}{100}$

$\therefore \varepsilon = 1.2V$

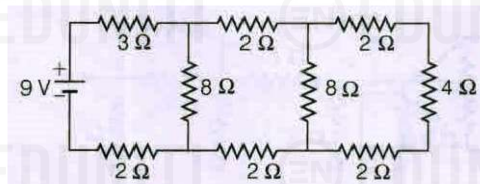
21.(a) From graph, I = 40 mA when l = 0.
In this condition, $\varepsilon = IR$ $\Rightarrow 1.2 = 0.04R$
$\therefore R = 30\Omega.$

22.(c) When l = 100 cm,

$I = \frac{\varepsilon - 6}{R} = \frac{1.2-6}{30} = -0.16A = -160mA$

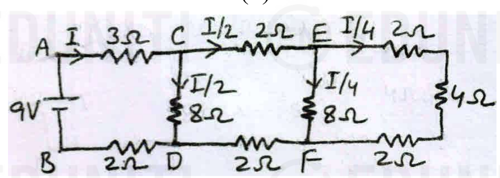
Its magnitude is 160 mA

23. In the circuit shown in the figure, the current through



(a) 3Ω resistor is 0.5 A (b) 3Ω resistor is 1 A
(c) 4 Ω resistor is 0.5 A (d) 4 Ω resistor is 0.25 A

23.(b,d)



$$R_{EF} = \frac{8 \times (2+4+2)}{8 + (2+4+2)} = \frac{8 \times 8}{8+8} = 4\Omega$$

$$R_{CD} = \frac{8 \times (2 + R_{EF} + 2)}{8 + (2 + R_{EF} + 2)} = \frac{8 \times 8}{8+8} = 4\Omega$$

$$R_{AB} = 3 + R_{CD} + 2 = 3 + 4 + 2 = 9\Omega$$

$$I_{3\Omega} = I = V / R_{AB} = 9/9 = 1A$$

$$I_{4\Omega} = I/4 = 0.25A$$

24. 24 cells, each of emf 1.5 V and internal resistance 2Ω are connected in mixed grouping so as to send the maximum current through a 12 Ω resistor. Then,
(a) there are two rows of 12 cells in each row
(b) the current in each row is 0.375 A
(c) the current in each row is 0.75 A
(d) the potential difference across 12 Ω resistor is 9 V

24.(a,b,d) Here $E = 1.5V$ is emf and $r = 2\Omega$ is internal resistance of each cell. Let there be m rows with n cells in each row. Then, mn = 24.

Then current through external resistance $R = 12\Omega$ is

$$i = \frac{nE}{R + nr/m} = \frac{mnE}{mR + nr} = \frac{24 \times 1.5}{12m + 2 \times 24/m} = \frac{3}{m + 4/m}$$

For i to be maximum, m + 4/m must be minimum,

or $\frac{d}{dm}\left(m + \frac{4}{m}\right) = 0$ or $1 - \frac{4}{m^2} = 0$ or $m = 2.$

Therefore, for i_{max}, m = 2 and n = 12. So, there are two rows of

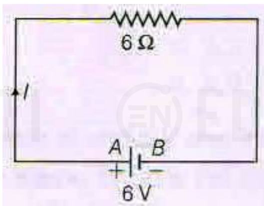
12 cells in each row. Also,

$$i_{\max} = \frac{3}{m+4/m} = \frac{3}{2+4/2} = 0.75A.$$

So, the current in each row is $i_{\max}/m = 0.375A.$

The p.d. across 12Ω resistor is $i_{\max}R = 0.75\times 12 = 9V.$

25. In the given circuit, AB is a wire of length 10 cm and resistance 6Ω. The number of electrons per unit volume in material of wire is 10^{29} m^{-3}. If cross-sectional area of wire is 1 mm^2, then which of the following options are correct



(a) Energy absorbed by electrons is 0.9×10^{-17} J
(b) Energy absorbed by electrons is 1.8×10^{-17} J
(c) Ohmic loss in wire is 6 J/s
(d) All kinetic energy of electrons will be lost in around 3.0×10^{-18} s

25.(b,c,d) Current in circuit, I = 6/6 = 1A

$$\Rightarrow v_d = \frac{I}{neA} = \frac{1}{10^{29}\times 1.6\times 10^{-19}\times 10^{-6}} = \frac{1}{1.6}\times 10^{-4}\,m/s$$

$\therefore$ Energy of electrons
= KE of one electron $\times$ No. of electrons

$$= \frac{1}{2}mv_d^2 \times (nAl)$$

$$= \frac{1}{2}\times 9.1\times 10^{-31}\times\left(\frac{1}{1.6}\times 10^{-4}\right)^2\times\left(10^{29}\times 10^{-6}\times 10^{-1}\right)$$

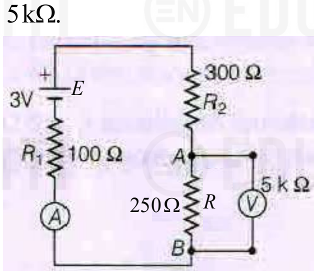
$$= 1.8\times 10^{-17}\text{ J}$$

Ohmic loss = $I^2R = (1)^2\times 6 = 6$ J/s

Time for loss of kinetic energy of all electrons

$$= \frac{1.8\times 10^{-17}}{6}s = 3\times 10^{-18}s$$

26. In given circuit, E_1 = 3V, $R_1 = 100\Omega$, $R_2 = 300\Omega$ and $R_3 = 250\Omega$. Assume cell and ammeter as ideal. The voltmeter resistance is $5k\Omega$.



(a) Voltmeter reads 3% lower value of potential drop across R_3
(b) Voltmeter reads 3% higher value of potential drop across R_3
(c) If V_{AB} is balanced against a potentiometer with potential gradient of 0.01 V/cm, then balance length is 115 cm
(d) If V_{AB} is balanced against a potentiometer with potential gradient of 0.01 V/cm, then balance length is 112 cm

26(a,c) In the absence of voltmeter,

$$V_{AB} = \frac{250}{250+(100+300)}\times 3 = \frac{15}{13}V$$

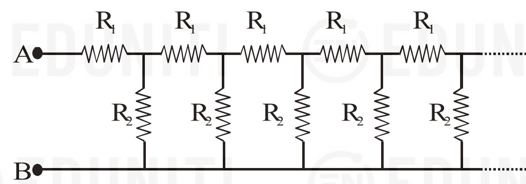
In the presence of voltmeter, $R_{AB} = \frac{250\times 5000}{250+5000} = \frac{5000}{21}\Omega$

and $V'_{AB} = \frac{5000/21}{5000/21+(100+300)}\times 3 = \frac{75}{67}V$

$$\therefore \text{ Fractional error} = \frac{75/67 - 15/13}{15/13} = -0.03 = -3\%$$

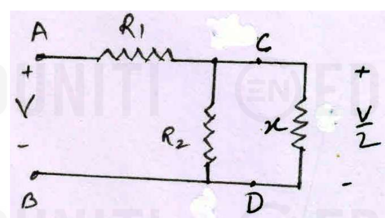
So, voltmeter reads 3% lower value. Balancing length of potentiometer wire, $l = \frac{V_{AB}}{k} = \frac{15/13}{0.01} = 115\,\text{cm}$

27. Consider an infinite ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage becomes successively half after each section. Then



(a) $\frac{R_1}{R_2} = 1$
(b) $\frac{R_1}{R_2} = \frac{1}{2}$
(c) $\frac{R_1}{R_2} = 2$
(d) $\frac{R_1}{R_2} = 3$

27.(a,c) Let $R_{AB} = x$. Then, $R_{CD} = x$



$$\Rightarrow x = R_1 + \frac{R_2x}{R_2+x} \qquad .....(1)$$

It is given that $V_{AB} = V$ and $V_{CD} = V/2$

So, p.d. across R_1 and R_2 are same i.e., $V/2$,

$$\Rightarrow R_1 = \frac{R_2x}{R_2+x} \qquad .....(2)$$

From (1) and (2) we get,

$$x = R_1 + R_1 = 2R_1 \text{ and } R_1 = \frac{R_2(2R_1)}{R_2+2R_1}$$

$$\Rightarrow R_1R_2 + 2R_1^2 = 2R_1R_2$$

$$\Rightarrow 2R_1^2 = R_1R_2 \qquad \therefore \frac{R_1}{R_2} = \frac{1}{2}$$

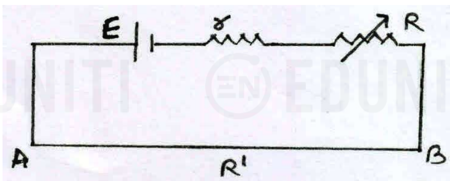
28. In a potentiometer wire experiment, the emf of a battery in the primary circuit is 20V and its internal resistance is 5Ω. There is a resistance box in series with the battery whose resistance can be varied from 120Ω to 170Ω. Resistance of the

potentiometer wire is 75Ω. Which of the following potential differences can be measured using this potentiometer?

(a) 5V (b) 6V
(c) 7V (d) 8V

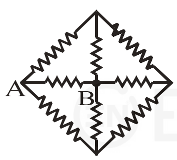
28.(a,b,c) Here, $E = 20V$, $r = 5\Omega$, $R' = 75\Omega$ and $R = 120\Omega$ to 170Ω

$$V_{AB} = \frac{ER'}{R'+r+R} = \frac{20\times 75}{75+5+R} = \frac{1500}{80+R}$$



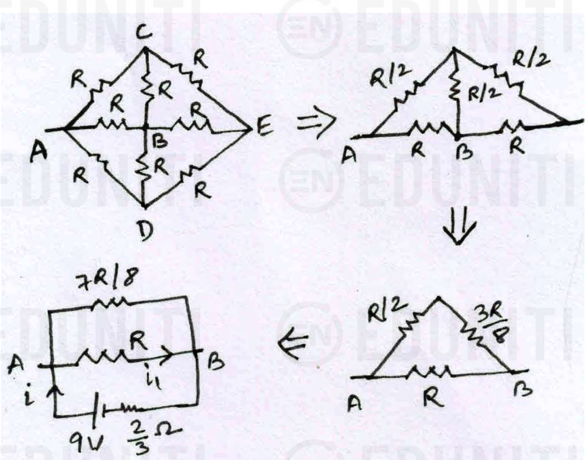
When $R = 120\Omega$, $V_{AB} = 7.5V$ and
when $R = 170\Omega$, $V_{AB} = 6V$
The maximum p.d. that can be measured across potentiometer wire is 7.5V. Practically, if we leave the points near the end of wire, the range of p.d. that can be measured is from slightly greater than zero to slightly less that 7.5V. Out of the given options, $5V$, $6V$ and $7V$ can be measured.

29. In the circuit shown, each resistor has resistance 5Ω. The points A and B are connected to the terminals of a cell of emf 9 volt and internal resistance $(2/3)\Omega$.



(a) The heat produced in the cell is 6W.
(b) The current in the resistor connected directly between A and B is 1.4A.
(c) The current in the resistor connected directly between A and B is 1.8 A.
(d) None of the above is correct.

29.(a,b)



As shown in figure, $R_{AB} = \dfrac{R\times 7R/8}{R+7R/8} = \dfrac{7R}{15}$

Here, $R = 5\Omega$ $\Rightarrow R_{AB} = 7/3\Omega$

$$\Rightarrow i = \frac{9}{R_{AB}+2/3} = \frac{9}{7/3+2/3} = 3A$$

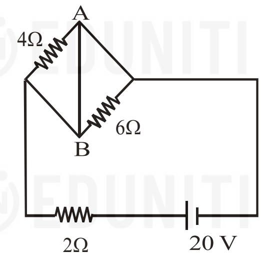
The heat produced in the cell is at the rate $i^2 r = 3^2 \times (2/3) = 6W$

$V_{AB} = iR_{AB} = 3\times 7/3 = 7V$

The current in the resistor connected directly between A and B is

$i_1 = V_{AB}/R = 7/5 = 1.4A$

30. In the circuit shown in figure,



(a) power supplied by the battery is 200 watt
(b) current flowing in the circuit is 5 A
(c) potential difference across $4\,\Omega$ resistance is equal to the potential difference across $6\,\Omega$ resistance
(d) current in wire AB is zero

30. (a,c)